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## DIRECT DRIVE GEAR REDUCER MOUNTING SYSTEMS

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### Abstract

A mounting system is provided that includes an electric motor, a gear reducer, a coupling, and a mounting bracket. The electric motor has an output shaft and a foot at a bottom of the electric motor. The gear reducer has an input shaft at a drive face. The coupling operatively connects the output shaft and the input shaft. The mounting bracket is secured to the drive face of the gear reducer and to the foot of the electric motor.

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims benefit of U.S. Provisional Application 63/556,486 filed Feb. 22, 2024, the entire contents of which are incorporated herein by reference.

### BACKGROUND

#### 1. Field of the Invention

[0002] The present disclosure is related to gear reducers. More particularly, the present disclosure is related to mounting systems for gear reducers.

#### 2. Description of Related Art

[0003] Industrial equipment such as, but not limited to, conveyors, packaging devices, manufacturing devices and others, commonly use electric motors and gear reducers to provide movement to various system components. In some instances, the gear reducers are directly mounted to the electric motor in what is commonly called a “direct drive reducer.” When using direct drive gear reducers, the mounting systems must be sufficient to transmit the torque from the motor through the gear reducer to the driven equipment.

[0004] A competing design constraint can be the need to minimize the size of the mounting system, which allows the system of the motor, reducer, and mounting system to fit within the available space of the end user.

[0005] While some mount systems for direct drive gear reducers are commercially available, it has been determined by the present disclosure that these solutions are not scalable to larger sized installations. It has also been determined by the present disclosure that even if the prior art mount systems were scalable, these systems fail to provide sufficient stress and/or thermal management of the drive train when in use.

[0006] Accordingly, there is a need for mounting systems that overcome, alleviate, and/or mitigate one or more of the aforementioned and other deleterious effects of the prior art.

### SUMMARY

[0007] A mounting system is provided that includes an electric motor, a gear reducer, a coupling, and a mounting bracket. The electric motor has an output shaft and a foot at a bottom of the electric motor. The gear reducer has an input shaft at a drive face. The coupling operatively connects the output shaft and the input shaft. The mounting bracket is secured to the drive face of the gear reducer and to the foot of the electric motor.

[0008] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes a tie-rod secured to the mounting bracket on a side opposite the electric motor.

[0009] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting bracket includes one or more lifting points.

[0010] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes an I-beam mount secured to a reducer foot at a bottom of the gear reducer.

[0011] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes a tie-rod secured to the I-beam mount on a side opposite the gear reducer.

[0012] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes a fan operatively secured to the electric motor.

[0013] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the fan draws in air axially across the gear reducer towards

the electric motor and forces the air radially outward.

[0014] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the gear reducer has a secondary shaft driving an oil pump. The oil pump places the gear reducer in fluid communication with an oil cooler. The oil cooler is positioned so that the air is forced across the oil cooler.

[0015] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes a shroud between the electric motor and the gear reducer so as to cover the output shaft, the input shaft, the coupling, and the fan.

[0016] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes a shroud between the electric motor and the gear reducer so as to cover the output shaft, the input shaft, and the coupling.

[0017] A mounting system is also provided that has an electric motor, a gear reducer, a coupling, a fan, and a shroud. The electric motor has an output shaft. The gear reducer has an input shaft at a drive face. The coupling operatively connects the output shaft to the input shaft. The fan is operatively secured to the electric motor. The shroud is between the electric motor and the gear reducer to cover the output shaft, the input shaft, the coupling, and the fan. The fan draws in air axially across the gear reducer and forces the air towards the electric motor where the air exits the shroud radially outward.

[0018] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the gear reducer includes a secondary shaft driving an oil pump. The oil pump places the gear reducer in fluid communication with an oil cooler. The oil cooler is positioned so that the air is forced across the oil cooler by the fan.

[0019] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes a mounting bracket secured to the drive face of the gear reducer and to a foot at the bottom of the electric motor.

[0020] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting bracket has one or more lifting points.

[0021] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes a tie-rod secured to the mounting bracket on a side opposite the electric motor.

[0022] In some embodiments either alone or together with any one or more of the aforementioned and/or after-mentioned embodiments, the mounting system further includes an I-beam mount secured to a reducer foot at a bottom of the gear reducer.

[0023] The above-described and other features and advantages of the present disclosure will be appreciated and understood by those skilled in the art from the following detailed description, drawings, and appended claims.

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## Description

### DESCRIPTION OF THE DRAWINGS

[0024] This application contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

[0025] FIG. 1 is a top perspective view of an exemplary embodiment of a mounting system according to the present disclosure;

[0026] FIG. 2 is a first side view of the mounting system of FIG. 1;

[0027] FIG. 3 is a second, opposite side view of the mounting system of FIG. 1;

[0028] FIG. 4 is another first side view of the mounting system of FIG. 1 having a shroud removed;

[0029] FIG. **5A** is a disassembled view of the mounting system of FIG. **1**;  
[0030] FIG. **5B** is a disassembled view of a gear reducer according to the present disclosure;  
[0031] FIG. **6** is a top perspective view of a shroud according to an exemplary embodiment of the present disclosure;  
[0032] FIG. **7** is a disassembled view of the shroud of FIG. **6**;  
[0033] FIG. **8** is a rear view of the shroud of FIG. **6** having the shroud illustrated in phantom;  
[0034] FIG. **9A** is a side view of the gear reducer of FIG. **1** shown having four testing locations;  
[0035] FIG. **9B** is a sectional view of the mounting system of FIG. **1** taken at slice **3** of FIG. **9A**;  
[0036] FIGS. **10A-13A** are air velocity simulations taken at the testing locations of FIG. **9A** for a comparison shroud;  
[0037] FIGS. **10B-13B** are air velocity simulations taken at the testing locations of FIG. **9A** for the shroud of the present disclosure;  
[0038] FIG. **14A** is a perspective view of a heat transfer simulation for a comparison shroud;  
[0039] FIG. **14B** is a perspective view of a heat transfer simulation for the shroud of the present disclosure;  
[0040] FIG. **15** is a perspective view of a mounting bracket according to an exemplary embodiment of the present disclosure shown secured to a gear reducer;  
[0041] FIG. **16** is a disassembled view the mounting bracket of FIG. **15**;  
[0042] FIG. **17** is a sectional view of the mounting bracket of FIG. **16** taken along lines **17-17**;  
[0043] FIG. **18** is a bottom perspective view of the mounting bracket in use with a foot mount;  
[0044] FIG. **19** is a bottom perspective view of the mounting system of FIG. **1** in use with an I-beam and I-beam foot mount;  
[0045] FIG. **20** is a top perspective view of the I-beam and I-beam foot mount of FIG. **19**;  
[0046] FIG. **21** is a top perspective view of an exemplary embodiment of a mounting bracket according to the present disclosure;  
[0047] FIG. **22** is a top view of the mounting bracket of FIG. **21**;  
[0048] FIG. **23** is a bottom view of the mounting bracket of FIG. **21**;  
[0049] FIG. **24** is a first side view of the mounting bracket of FIG. **21**;  
[0050] FIG. **25** is a second, opposite side view of the mounting bracket of FIG. **21**;  
[0051] FIG. **26** is a first end view of the mounting bracket of FIG. **21**;  
[0052] FIG. **27** is a second, opposite end view of the mounting bracket of FIG. **21**;  
[0053] FIG. **28** is a top perspective view of an exemplary embodiment of a shroud according to the present disclosure;  
[0054] FIG. **29** is a top view of the shroud of FIG. **28**;  
[0055] FIG. **30** is a bottom view of the shroud of FIG. **28**;  
[0056] FIG. **31** is a first side view of the shroud of FIG. **28**;  
[0057] FIG. **32** is a second, opposite side view of the shroud of FIG. **28**;  
[0058] FIG. **33** is a first end view of the shroud of FIG. **28**; and  
[0059] FIG. **34** is a second, opposite end view of the shroud of FIG. **28**

#### DETAILED DESCRIPTION

[0060] Referring to the drawings and in particular to FIGS. **1-5B**, an exemplary embodiment of a mounting system according to the present disclosure is shown and is generally referred to by reference number **10**. Advantageously, mounting system **10** is configured to secure an electric motor **12** and a direct drive gear reducer **14** to one another in a structurally stable and compact manner.

[0061] Additionally, mounting system **10** can, in some embodiments, include a cooling fan **16** that is driven by motor **12** to draw in ambient air and force the air across gear reducer **14**, which pulls heat out of the gear reducer. In some embodiments, fan **16** is configured to draw air into system **10** axially across gear reducer **14**, push the air towards motor **12**, and to exhaust the air radially outward from shroud **18**. Of course, it is contemplated by the present disclosure for fan **16** to have

any desired configuration such as, but not limited to, drawing air in into shroud **18** radially at motor **12** and forcing the air axially towards gear reducer **14**.

[0062] Motor **12** can be any electric machine having sufficient power to drive a driven device (not shown) operatively connected to gear reducer **14**. In some embodiments, motor **12** is an electric motor having a horsepower rating of between **20** to **400**, preferably between **40** and **340**, more preferably between **60** and **300**, and any subranges therebetween.

[0063] Reducer **14** can be any direct drive gear reducer. In the illustrated embodiment, gear reducer **14** is a beltless reducer having a split housing **20, 22** having cooling fins **24** defined on an exterior. Fins **24** are, preferably, positioned on housing **20, 22** in an axial direction, namely in a direction that is parallel to a direction of the flow of air drawn in by fan **16**.

[0064] In some embodiments, gear reducer **14** can include a secondary shaft **26**, which system **10** advantageously is configured to use to drive an oil pump **28** that is in fluid communication with an oil cooler **30**. Oil cooler **30** is positioned so that air that is drawn in to shroud **18** by fan **16** passes through and exchanges heat with the oil in the oil cooler. Simply stated, pump **28** pumps heated oil from gear reducer **14** through cooler **30**, where heat in the oil is removed by fan **16** and returns the oil to the gear reducer.

[0065] System **10** includes a mounting bracket **32** that is mounted to one or more feet **34** of motor **12** and mounted to a drive face **36** of gear reducer **14**. Thus, bracket **32** provides a structural member that is secured to drive face **36** in a cantilevered manner so that the bracket is secured to feet **34** of motor **12**. Bracket **32** is described in more detail with reference to FIGS. **15-18**.

[0066] Advantageously, system **10**, due at least in part to bracket **32** being secured to drive face **36**, provides a structurally secure connection of motor **12** and gear reducer **14** in a simple and structurally secure manner that reduces the space necessary to secure the components to one another as compared to prior art connections.

[0067] In some embodiments, bracket **32** can include one or more lifting points **38** integrally formed therewith.

[0068] Bracket **32** is configured to provide a rigid base between motor **12** and gear reducer **14**, reducing stress to the shafts and bearings in system **10**. In some embodiments, bracket **32** can be a cast metallic support. Bracket **32** can have a cross section with gussets **40** and/or C-shaped bottom rails **42**, which provide the bracket with sufficient section modulus to resist the applied bending moment. In some embodiments, gussets **40** are angled to provide sufficient space to receive a coupling **44** therebetween.

[0069] Further, bracket **32** provides a closed bottom to the area enclosed by shroud **18** to improve the flow of air through the shroud due to fan **16**.

[0070] System **10** further includes coupling **44** that operatively connects an output shaft **46** of motor **12** to an input shaft **48** of gear reducer **14**. In some embodiments, coupling **44** can provide one or more degrees of freedom between output and input shafts **46, 48**. The one or more degrees of freedom can compensate for tolerances such as, but not limited to, runout of the output/input shafts **46, 48**, which can transmit damaging radial loads into bearings of gear reducer **14**.

[0071] In one embodiment, coupling **44** is an elastomeric coupling such as those commercially available from Applicant and known as the Dodge Raptor coupling.

[0072] Moreover, system **10** leaves feet **50** of gear reducer **14** free from the connection with motor **12**, which provides the ability for system **10** to, in some embodiments, a further support shown as an I-beam mount **52**. Here, feet **40** can be machined to a flat plane so as to ensure a secure connection with I-beam mount **52**.

[0073] System **10** can include one or more tie-rods **54** that secure bracket **32** or I-beam mount **52** to a rigid mounting location (not shown). In the illustrated embodiment, bracket **32** is configured so that tie-rod **54** is secured to a bottom of the bracket at an approximate location of a center of gravity of motor **12**.

[0074] In the illustrated embodiment, fan **16** is operatively connected to output shaft **46** of motor

**12.** Of course, it is contemplated by the present disclosure for fan **16** to be operatively connected to input shaft **48** of gear reducer **14**, to coupling **44**, and any combinations thereof. Preferably, fan **16** is configured to draw in ambient air into shroud **18** axially across gear reducer **14**, force the air axially along shafts **46**, **48** within the shroud towards motor **12**, and then radially out of shroud **18**. [0075] Referring now to FIGS. **20-21**, I-beam mount **52** is described in more detail. Advantageously, I-beam mount **52** provides easy access to mechanical fasteners used to secure the mount to gear reducer **14**. In some embodiments, best seen in FIG. **20**, I-beam mount **52** includes a rear set of fastener openings **56** and a plurality of front sets of fastener openings **58**. [0076] The rear fastener openings **56** are securable to the rear feet **50** of gear reducer **14**, while the front sets of fastener openings **58** can be selectively connected to the front feet **50** of reducer. In this manner, I-beam mount **52** can be used for connection to reducers **14** of different sizes—eliminating parts and increasing the compatibility of system **10**. Here, the moment arm (i.e., the distance) between front fastener openings **58** and tie-rod **54** is reduced as the size of reducer and, thus, the torque applied to I-beam mount **52**, increases. [0077] Moreover, I-beam mount **52** serves as an extension for mounting the torque arm further away from the gearbox output moment than would otherwise be possible. Simply, I-beam mount **52** is configured, in some embodiments, so that front sets of fastener openings **58** are located at an axial direction past drive face **36** of gear reducer **14**. [0078] Reducer **14** is configured so that rear and front feet **50** are spaced from one another so that there is no overlap of the front openings **58** in I-beam **52**, which allows the same I-beam to be used for multiple reducer sizes. [0079] Shroud **18** is described in more detail with reference to FIGS. **6-14B**. [0080] Shroud **18** includes a first end **60** at motor **12**, a second end **62** at gear reducer **14**, and an open bottom end **64**. Shroud **18** is secured between motor **12** and gear reducer **14** with open end **64** connected to bracket **32** in a manner sufficient to cover fan **16**, output shaft **46**, input shaft **48**, coupling **44**, and, when present, oil cooler **30**. In this manner, shroud **32** can prevent inadvertent contact with the moving parts of system **10**. [0081] In addition, shroud **18** is advantageously configured to maximize the cooling efficiency of fan **16** by reducing ingress of air except through the inlet formed between second end **62** and gear reducer **14** and/or by reducing leakage of air except through the radial vents in first end **60**. [0082] Shroud **18** can include a fan cover region **66**. Region **66** includes a plurality of air vents **68** at its outer extents so that air drawn in by fan **16** at second end **62** proximate gear reducer **14** is pulled axially through shroud **18** and forced radially outward from region **66** through the air vents **68**. Without wishing to be bound by any particular theory, it is believed that exhausting of air drawn in by fan **16** radially through air vents **68** minimizes air flow restrictions through shroud **18**, which increases the cooling capacity of system **10**. [0083] Region **66** includes a shaft slot **70** that permits shroud **18** to be positioned over output shaft **46** with bottom end **64** on bracket **32** at least at first end **60**. In some embodiments, shroud **18** can include a slot cover **72** that can be connected to fan cover region **66** to block off shaft slot **70**. Again and without wishing to be bound by any particular theory, it is believed that blocking off shaft slot **70** via cover **72** reduces air leakage at first end **60** proximate motor **12** axially outward and reduces suction of air axially into the first end—forcing the air to be drawn in at second end **62** proximate gear reducer **14**. [0084] Shroud **18** includes a narrowed region **74** prior, in the direction of airflow, to fan cover region **66**. Again and without wishing to be bound by any particular theory, it is believed that fan cover region **66** having a larger outer dimension than narrowed region **74** provide enhanced cooling capacity. [0085] For example, fan cover region **66** provides a substantially uniform suction area around fan **16** which can increase the efficiency of the fan **16**. As another example, the narrowed region **74** provides an area where the air is accelerated as it is drawn into fan cover region **66**.

[0086] In some embodiments, shroud **18** includes connectors **76** that connect bottom end **64** to bracket **32** at least in narrowed region **74**. Connectors **76** are believed to reduce ingress of air except through the inlet formed between second end **62** and gear reducer **14** and/or by reducing leakage of air between bottom end **64** and bracket **32**.

[0087] Shroud **18** further includes an air inlet region **78** at second end **62** proximate gear reducer **14**. Inlet region **78** has a larger outer dimension than narrowed region **74**. In some embodiments, shroud **18** includes a tapered zone **80** between narrowed region **74** and inlet region **78** to eliminate sharp turns in airflow. In still other embodiments, shroud **18** can include radial extensions **82** at second end **62**. Extensions **82** are believed to ensure that the ingress of air through inlet region **78** occurs as close to gear reducer **14** as possible.

[0088] In some embodiments, air inlet region **78** is sized and configured to provide sufficient space to receive coupling **44** with sufficient degrees of freedom between shafts **46**, **48**.

[0089] In some embodiments, shroud **18** is configured so that narrowed region **74**, air inlet region **78**, and, when present, tapered zone **80** have a rounded or oval cross section. Without wishing to be bound by any particular theory, it is believed that the rounded cross section of shroud **18** at least in regions **74**, **78**, and **80** provides an increased air flow as compared to other configurations.

[0090] FIGS. **9A-14B** illustrate air velocity and heat transfer capability testing performed using computational fluid dynamics (CFD) simulations.

[0091] FIG. **9A** indicates the four locations or slices on which air velocity CFD simulations were performed. Slices **1** and **2** are taken within shroud **18**, while slice **3** is taken at second end **62** of the shroud, namely at the location where air is drawn into the shroud. Slice **4** is taken at a location prior to second end **62** of shroud **18**.

[0092] The air velocity simulation for a comparison shroud having an octagonal shape are shown in FIGS. **10A-13A**, while the air velocity simulation for shroud **18** are shown in FIGS. **10B-13B**.

[0093] Comparing FIGS. **10A** to **10B**, taken at a location where the comparison shroud and shroud **18** are not present, the air velocities are substantially similar.

[0094] Comparing FIGS. **11A** to **11B**, taken at the second end **62** or air inlet location for the comparison shroud and shroud **18**, shroud **18** provides larger air velocities particularly in the upper regions of gear reducer **14**.

[0095] Comparing FIGS. **12A** to **12B** and **13A** to **13B**, shroud **18** provides larger air velocities in the upper regions and lower regions of gear reducer **14**.

[0096] The heat transfer capability tests performed using computational fluid dynamics (CFD) simulations are shown in FIGS. **14A** and **14B**. Here, FIG. **14A** shows the heat transfer capability for the comparison shroud having an octagonal shape, while FIG. **14B** shows the heat transfer capability for shroud **18**. Here, the heat transfer capability of shroud **18** is larger in both the central and upper regions of gear reducer **14**.

[0097] Turning now to FIGS. **21-27**, an exemplary embodiment of a design for a mounting bracket **32** according to the present disclosure is shown. Additionally and with reference to FIGS. **28-34**, an exemplary embodiment of a design for a shroud **18** according to the present disclosure is shown.

[0098] It should also be noted that the terms “first”, “second”, “third”, “upper”, “lower”, and the like may be used herein to modify various elements. These modifiers do not imply a spatial, sequential, or hierarchical order to the modified elements unless specifically stated.

[0099] While the present disclosure has been described with reference to one or more exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the disclosure without departing from the scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment(s) disclosed as the best mode contemplated, but that the disclosure will include all embodiments falling within the scope of the appended claims.

## **PARTS LIST**

[0100] mounting system **10** output shaft **46** [0101] input shaft **48** electric motor **12** [0102] feet **50**  
gear reducer **14** [0103] I-beam mount **52** cooling fan **16** [0104] tie-rods **54** shroud **18** [0105] rear  
fastener openings **56** housing **20** [0106] front fastener openings **58** housing **22** [0107] first end **60**  
cooling fins **24** [0108] second end **62** secondary shaft **26** [0109] open bottom end **64** oil pump **28**  
[0110] fan cover region **66** oil cooler **30** [0111] bracket **32** air vents **68** [0112] feet **34** shaft slot **70**  
[0113] drive face **36** slot cover **72** [0114] lifting points **38** narrowed region **74** [0115] gussets **40**  
connectors **76** [0116] bottom rail **42** air inlet region **78** [0117] coupling **44** tapered zone **80** [0118]  
radial extensions **82**

## **Claims**

- 1.** A mounting system comprising: an electric motor having an output shaft and a foot at a bottom of the electric motor; a gear reducer having an input shaft at a drive face; a coupling operatively connecting the output shaft to the input shaft; and a mounting bracket secured to the drive face of the gear reducer and to the foot of the electric motor.
- 2.** The mounting system of claim 1, further comprising a tie-rod secured to the mounting bracket on a side opposite the electric motor.
- 3.** The mounting system of claim 1, wherein the mounting bracket comprises one or more lifting points.
- 4.** The mounting system of claim 1, further comprising an I-beam mount secured to a reducer foot at a bottom of the gear reducer.
- 5.** The mounting system of claim 4, further comprising a tie-rod secured to the I-beam mount on a side opposite the gear reducer.
- 6.** The mounting system of claim 1, further comprising a fan operatively secured to the electric motor.
- 7.** The mounting system of claim 6, wherein the fan draws in air axially across the gear reducer towards the electric motor and forces the air radially outward.
- 8.** The mounting system of claim 7, wherein the gear reducer further comprises a secondary shaft driving an oil pump, the oil pump placing the gear reducer in fluid communication with an oil cooler, and wherein the oil cooler is positioned so that the air is forced across the oil cooler.
- 9.** The mounting system of claim 6, further comprising a shroud between the electric motor and the gear reducer so as to cover the output shaft, the input shaft, the coupling, and the fan.
- 10.** The mounting system of claim 1, further comprising a shroud between the electric motor and the gear reducer so as to cover the output shaft, the input shaft, and the coupling.
- 11.** A mounting system comprising: an electric motor having an output shaft; a gear reducer having an input shaft at a drive face; a coupling operatively connecting the output shaft to the input shaft; a fan operatively secured to the electric motor; and a shroud between the electric motor and the gear reducer so as to cover the output shaft, the input shaft, the coupling, and the fan, wherein the fan draws in air axially across the gear reducer and forces the air towards the electric motor where the air exits the shroud radially outward.
- 12.** The mounting system of claim 11, wherein the gear reducer further comprises a secondary shaft driving an oil pump, the oil pump placing the gear reducer in fluid communication with an oil cooler, and wherein the oil cooler is positioned so that the air is forced across the oil cooler by the fan.
- 13.** The mounting system of claim 11, further comprising a mounting bracket secured to the drive face of the gear reducer and to a foot at the bottom of the electric motor.
- 14.** The mounting system of claim 13, wherein the mounting bracket comprises one or more lifting points.
- 15.** The mounting system of claim 13, further comprising a tie-rod secured to the mounting bracket



on a side opposite the electric motor.

**16.** The mounting system of claim 13, further comprising an I-beam mount secured to a reducer foot at a bottom of the gear reducer.

**17.** The mounting system of claim 16, further comprising an I-beam mount secured to a reducer foot at a bottom of the gear reducer.

**18.** The mounting system of claim 11, further comprising a tie-rod secured to the I-beam mount on a side opposite the gear reducer.

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